

Project Design Phase-II

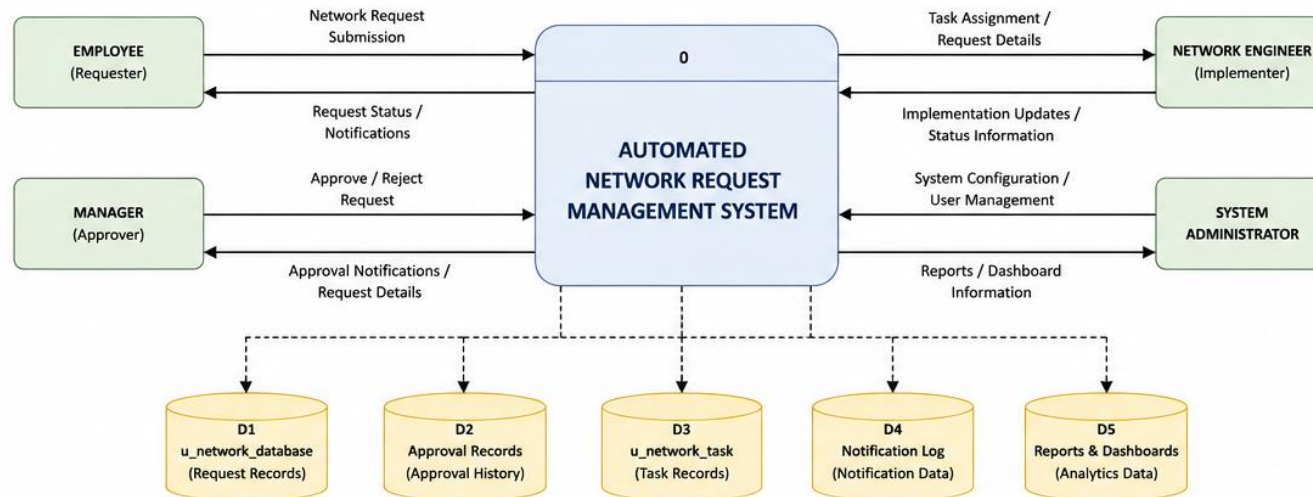
Data Flow Diagram & User Stories

Date	26 June 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

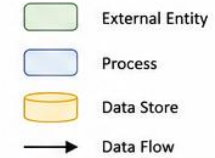
Data Flow Diagrams: A **Data Flow Diagram (DFD)** is a graphical representation of the flow of information within the **Automated Network Request Management System**. It illustrates how network service requests are submitted by users through the ServiceNow Service Portal, processed by the system, validated, approved, assigned to network engineers, and fulfilled. The DFD clearly depicts how data enters the system, how it is transformed through various processes, where it is stored in the custom tables (u_network_database and u_network_task), and how it flows between employees, managers, network engineers, and the ServiceNow platform. A well-designed DFD provides a clear understanding of the system's functionality, improves communication among stakeholders, and serves as an effective tool for analyzing, designing, and documenting the automated network request management process.

DATA FLOW DIAGRAM (DFD)

LEVEL – 0 DFD (CONTEXT DIAGRAM)



LEGEND



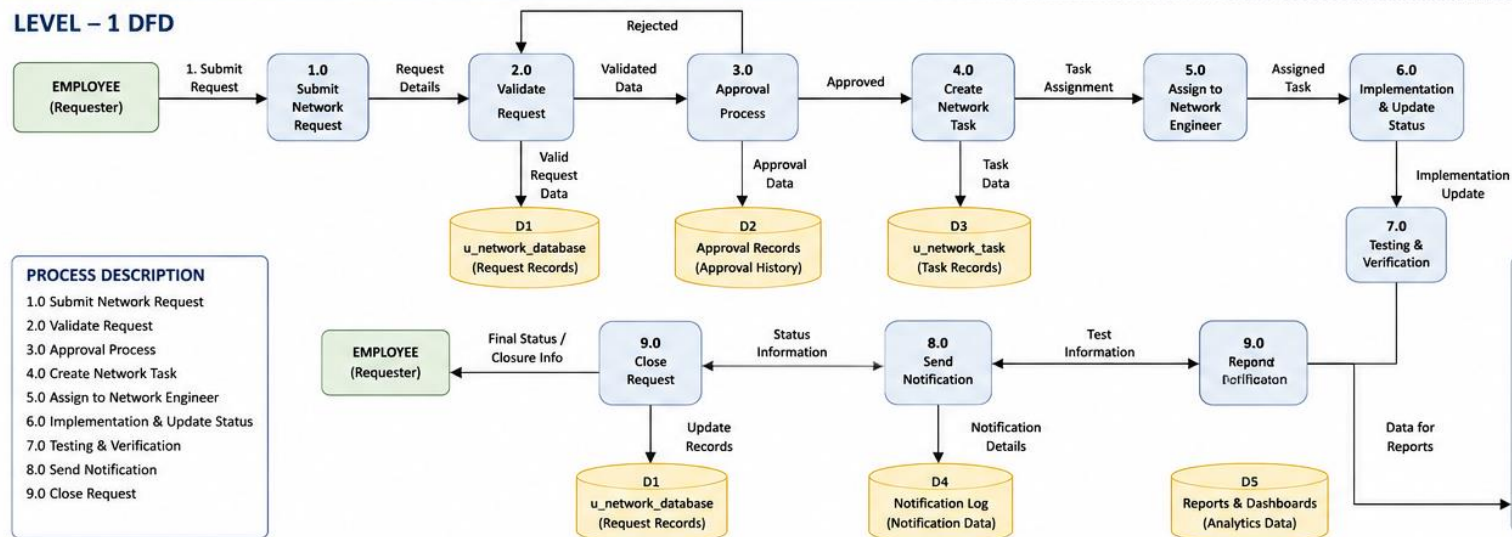
EXTERNAL ENTITIES

- Employee (Requester)
- Manager (Approver)
- Network Engineer (Implementer)
- System Administrator

DATA STORES

- D1 – u_network_database
- D2 – Approval Records
- D3 – u_network_task
- D4 – Notification Log
- D5 – Reports & Dashboards

LEVEL – 1 DFD



PROCESS DESCRIPTION

- 1.0 Submit Network Request
- 2.0 Validate Request
- 3.0 Approval Process
- 4.0 Create Network Task
- 5.0 Assign to Network Engineer
- 6.0 Implementation & Update Status
- 7.0 Testing & Verification
- 8.0 Send Notification
- 9.0 Close Request

DATA FLOW SUMMARY

1. Employee submits network request.
2. System validates the request and stores in u_network_database.
3. Approval workflow processes the request.
4. On approval, task is created in u_network_task and assigned.
5. Engineer implements and updates status.
6. Testing & verification is completed.
7. Notifications are sent to stakeholders.
8. Request is closed and records updated.
9. Data is used for reports & dashboards.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee (Requester)	User Authentication	USN-1	As an employee, I can log in to the ServiceNow portal using my authorized credentials to access network services.	I can successfully access the Service Portal.	High	Sprint-1
	Network Request Submission	USN-2	As an employee, I can submit a network service request by selecting the required service and entering the necessary details.	I can create and submit a network request successfully.	High	Sprint-1
	Request Tracking	USN-3	As an employee, I can track the status of my submitted network requests in real time.	I can view the current request status and progress.	High	Sprint-2
	Email Notifications	USN-4	As an employee, I want to receive email notifications whenever my request status changes.	I receive notifications for request submission, approval, rejection, and completion.	High	Sprint-2
	Request History	USN-5	As an employee, I can view all my previously submitted network requests.	I can access my request history from the portal.	Medium	Sprint-3
Manager (Approver)	Request Approval	USN-6	As a manager, I can review network requests and approve or reject them based on organizational policies.	I can approve or reject pending requests.	High	Sprint-2
	Approval Comments	USN-7	As a manager, I can provide comments while approving or rejecting a request.	Comments are saved with the approval record.	Medium	Sprint-2
Network Engineer	Task Management	USN-8	As a network engineer, I can receive automatically assigned implementation tasks after request approval.	I can view assigned network tasks in my work queue.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
System Administrator	Request Implementation	USN-9	As a network engineer, I can update task progress and implementation details until completion.	Task status and implementation notes are successfully updated.	High	Sprint-3
	Workflow Automation	USN-10	As a system administrator, I can configure workflow automation, approvals, and assignment rules for network requests.	The system automatically routes requests according to configured workflows.	High	Sprint-3
	Dashboard & Reports	USN-11	As a system administrator, I can generate reports and dashboards to monitor network request performance.	Reports and dashboards display accurate request metrics and analytics.	Medium	Sprint-3
	Role-Based Access Control	USN-12	As a system administrator, I can manage user roles and permissions to ensure secure access to the application.	Users can only access modules and records based on their assigned roles.	Medium	Sprint-3